```
%%shell
apt-get update && apt-get install -y gcc
```

Show hidden output

Assignment 1

```
%%writefile a1.c

#include<stdio.h>
void main()
{
int year;
printf("Enter year: ");
scanf("%d",&year);
if ((year%4==0 && year%100!=0) ||(year%400==0))
{
printf("%d is a leap year. \n", year);
}
else
{
printf("%d is not a leap year. \n", year);
}
}
```

```
Writing a1.c
```

```
%%shell
gcc a1.c
./a.out
```

```
Enter year: 2024
2024 is a leap year.
```

Assignment 2

```
%%writefile a2.c

#include <stdio.h>
void main()
{
int a, b, choice;
float r;
printf("SIMPLE CALCULATOR: \n 1:Addition \n 2:Subtraction \n 3:Multiplication \n 4:Division \n");
printf("Enter Choice: ");
scanf("%d", &choice);
printf("Enter two numbers: ");
scanf("%d %d", &a, &b);
switch(choice){
case 1:
{
r=a+b;
printf("%f", r);
}
break;
case 2:
{
r=a-b;
printf("%f", r);
}
break;
case 3:
{
r=a*b;
printf("%f", r);
}
break;
case 4:
{
if (b==0)
{
printf("Divisor cannot be 0");
}
else
{
r=(float)a/b;
```

```
printf("%f", r);
}
}
break;
default:
printf("Error:Invalid Choice");
}
}
```

```
Writing a2.c
```

```
%%shell
gcc a2.c
./a.out
```

```
SIMPLE CALCULATOR:
 1:Addition
 2:Subtraction
 3:Multiplication
 4:Division
Enter Choice: 2
Enter two numbers: 25 4
21.000000
```

Assignment 3

```
%%writefile a3.c

#include<stdio.h>
void main()
{
int n, r, i=1;
printf("Enter number: ");
scanf("%d", &n);
while (i<=10)
{
r=n*i;
printf("%d \n", r);
i++;
}
}
```

```
Writing a3.c
```

```
%%shell
gcc a3.c
./a.out
```

```
Enter number: 5
5
10
15
20
25
30
35
40
45
50
```

Assignment 4

```
%%writefile a4.c

#include <stdio.h>

int main() {
    int bp;
    float hra, ta, gs, pt, ns;
    printf("Enter Basic Pay: ");
    scanf("%d",&bp);
    hra=0.1*bp;
    ta=0.05*bp;
    gs=bp+hra+ta;
    printf("Gross Salary: %f\n", gs);
    pt=0.02*gs;
    ns=gs-pt;
    printf("Net Salary:%f", ns);
```

```
    return 0;
}
```

```
Overwriting a4.c
```

```
%%shell
gcc a4.c
./a.out
```

```
Enter Basic Pay: 500000
Gross Salary: 575000.000000
Net Salary:563500.000000
```

Assignment 5

```
%%writefile a5.c

#include <stdio.h>

void mat_add(int r, int c, int a[10][10], int b[10][10]) {
    printf("\nAddition:\n");
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++)
            printf("%d ", a[i][j] + b[i][j]);
        printf("\n");
    }
}

void get_saddle(int r, int c, int m[10][10]) {
    int f = 0;
    for (int i = 0; i < r; i++) {
        int min = m[i][0], c_idx = 0;
        for (int j = 1; j < c; j++) {
            if (m[i][j] < min) { min = m[i][j]; c_idx = j; }
        }
        int k;
        for (k = 0; k < r; k++)
            if (m[k][c_idx] > min) break;
        if (k == r) { printf("\nSaddle point: %d", min); f = 1; }
    }
    if (!f) printf("\nNo saddle point.");
}

void is_magic(int n, int m[10][10]) {
    int d1 = 0, d2 = 0;
    for (int i = 0; i < n; i++) {
        d1 += m[i][i]; d2 += m[i][n-1-i];
    }
    if (d1 != d2) { printf("\nNot magic."); return; }
    for (int i = 0; i < n; i++) {
        int rs = 0, cs = 0;
        for (int j = 0; j < n; j++) { rs += m[i][j]; cs += m[j][i]; }
        if (rs != d1 || cs != d1) { printf("\nNot magic."); return; }
    }
    printf("\nMagic square. Sum: %d", d1);
}

void mat_inv(int m[10][10]) {
    float det = (m[0][0] * m[1][1]) - (m[0][1] * m[1][0]);
    if (det == 0) { printf("\nInverse doesn't exist."); return; }
    printf("\nInverse (2x2):\n");
    printf("%.2f %.2f\n", m[1][1]/det, -m[0][1]/det);
    printf("%.2f %.2f\n", -m[1][0]/det, m[0][0]/det);
}

int main() {
    int r, c, m1[10][10], m2[10][10];
    printf("Rows/Cols: ");
    scanf("%d %d", &r, &c);
    printf("Mat A:\n");
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++) scanf("%d", &m1[i][j]);
    printf("Mat B:\n");
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++) scanf("%d", &m2[i][j]);

    mat_add(r, c, m1, m2);
    get_saddle(r, c, m1);
    if (r == c) is_magic(r, m1);
    if (r == 2 && c == 2) mat_inv(m1);
    else printf("\nInverse only for 2x2 in this code.");
```

```
    printf("\n");
    return 0;
}
```

```
Overwriting a5.c
```

```
%%shell
gcc a5.c
./a.out
```

```
Rows/Cols: 2 2
Mat A:
3 4
5 7
Mat B:
2 5
5 8

Addition:
5 9
10 15

Saddle point: 5
Not magic.
Inverse (2x2):
7.00 -4.00
-5.00 3.00
```

Assignment 6

```
%%writefile a6a.c

#include <stdio.h>

long fact_iter(int n) {
    long res = 1;
    for (int i = 2; i <= n; i++) {
        res *= i;
    }
    return res;
}

int main() {
    int num;
    printf("Enter number: ");
    scanf("%d", &num);
    if (num < 0) {
        printf("Invalid input.\n");
    } else {
        printf("Factorial : %ld\n", fact_iter(num));
    }
    return 0;
}
```

```
Writing a6a.c
```

```
%%shell
gcc a6a.c
./a.out
```

```
Enter number: 4
Factorial : 24
```

```
%%writefile a6b.c

#include <stdio.h>

long fact_rec(int n) {
    if (n <= 1) return 1;
    return n * fact_rec(n - 1);
}

int main() {
    int num;
    printf("Enter number: ");
    scanf("%d", &num);
    if (num < 0) {
        printf("Invalid input.\n");
    } else {
```

```
        printf("Factorial : %ld\n", fact_rec(num));
    }
    return 0;
}
```

```
Writing a6b.c
```

```
%%shell
gcc a6b.c
./a.out
```

```
Enter number: 4
Factorial : 24
```

Assignment 7

```
%%writefile a7.c

#include <stdio.h>

struct student {
    int roll;
    char name[50];
    int marks[3];
    int total;
    float perc;
};

void calculate(struct student *s) {
    s->total = 0;
    for (int i = 0; i < 3; i++) {
        s->total += s->marks[i];
    }
    s->perc = s->total / 3.0;
}

int main() {
    struct student s[50];
    int n;

    printf("Enter number of students: ");
    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        printf("\nStudent %d Details:\n", i + 1);
        printf("Roll Number: ");
        scanf("%d", &s[i].roll);
        printf("Name: ");
        scanf("%s", s[i].name);
        printf("Marks in 3 subjects: ");
        for (int j = 0; j < 3; j++) {
            scanf("%d", &s[i].marks[j]);
        }
        calculate(&s[i]);
    }

    printf("\n--- Student Results ---\n");
    for (int i = 0; i < n; i++) {
        printf("\nRoll No: %d\n", s[i].roll);
        printf("Name: %s\n", s[i].name);
        printf("Total Marks: %d\n", s[i].total);
        printf("Percentage: %.2f%%\n", s[i].perc);
    }

    return 0;
}
```

```
Writing a7.c
```

```
%%shell
gcc a7.c
./a.out
```

```
Enter number of students: 3

Student 1 Details:
Roll Number: 4
Name: abc
Marks in 3 subjects: 23 43 55
```

```
Student 2 Details:
Roll Number: 3
Name: def
Marks in 3 subjects: 23
34 55

Student 3 Details:
Roll Number: 2
Name: ghi
Marks in 3 subjects: 54 56 34

--- Student Results ---

Roll No: 4
Name: abc
Total Marks: 121
Percentage: 40.33%

Roll No: 3
Name: def
Total Marks: 112
Percentage: 37.33%

Roll No: 2
Name: ghi
Total Marks: 144
Percentage: 48.00%
```

Assignment 8

```
%%writefile a8.c

#include <stdio.h>

int main() {
    int m1, m2, m3, m4, m5, T;
    float P;

    printf("Enter 5 marks: ");
    scanf("%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5);

    if (m1 < 40 || m2 < 40 || m3 < 40 || m4 < 40 || m5 < 40) {
        printf("FAIL\n");
    } else {
        T = m1 + m2 + m3 + m4 + m5;
        P = T / 5.0;

        printf("Total: %d, Percentage: %.2f%%\n", T, P);
        printf("Result: PASS - ");

        if (P >= 75) {
            printf("Distinction\n");
        } else if (P >= 60) { // No need to check P < 75, the 'else' already implies it
            printf("First Division\n");
        } else if (P >= 50) {
            printf("Second Division\n");
        } else {
            printf("Third Division\n");
        }
    }

    return 0;
}
```

```
Writing a8.c
```

```
%%shell
gcc a8.c
./a.out
```

```
Enter 5 marks: 92 56 82 89 78
Total: 397, Percentage: 79.40%
Result: PASS - Distinction
```

Assignment 9

```
%%writefile a9.c

#include <stdio.h>

int main() {
```

```
    int a, b, c, d=0;
    printf("Enter number: ");
    scanf("%d", &a);
    b=a;
    while(b>0)
    {
    c=b%10;
    d=d+(c*c*c);
    b=b/10;
    }
    if (a==d){
    printf("Armstrong Number");
    }
    else{
    printf("Not Armstrong Number");
    }
    return 0;
    }
```

```
Writing a9.c
```

```
%%shell
gcc a9.c
./a.out
```

```
Enter number: 407
Armstrong Number
```

Assignment 10

```
%%writefile a10.c

#include <stdio.h>
#include <math.h>

int main() {
    int a, b, n, choice, i;
    unsigned long long f = 1;
    float r;

    printf("SIMPLE CALCULATOR: \n 1:Addition \n 2:Subtraction \n 3:Multiplication \n 4:Division \n 5:Power \n 6:Factorial \
    printf("Enter Choice: ");
    scanf("%d", &choice);

    if (choice >= 1 && choice <= 5) {
        printf("Enter two numbers: ");
        scanf("%d %d", &a, &b);
    } else if (choice == 6) {
        printf("Enter number: ");
        scanf("%d", &n); // Read into n, not f
    }

    switch(choice) {
        case 1: r = a + b; printf("Result: %f\n", r); break;
        case 2: r = a - b; printf("Result: %f\n", r); break;
        case 3: r = a * b; printf("Result: %f\n", r); break;
        case 4:
            if (b == 0) printf("Divisor cannot be 0\n");
            else { r = (float)a / b; printf("Result: %f\n", r); }
            break;
        case 5: r = pow(a, b); printf("Result: %f\n", r); break;
        case 6:
            if (n < 0) printf("Error: Negative number\n");
            else {
                for (i = 1; i <= n; ++i) f *= i;
                printf("Factorial of %d = %llu\n", n, f);
            }
            break;
        default:
            printf("Error: Invalid Choice\n");
            break;
    } // Added missing closing brace for switch

    return 0;
}
```

```
Overwriting a10.c
```

```
!gcc a10.c -o calculator -lm
!./calculator
```

```
SIMPLE CALCULATOR:
 1:Addition
 2:Subtraction
 3:Multiplication
 4:Division
 5:Power
 6:Factorial
Enter Choice: 5
Enter two numbers: 2 3
Result: 8.000000
```

Assignment 11

```
%%writefile a11.c

#include <stdio.h>
#include <math.h>

void check_prime(int n) {
    int f = 0;
    if (n <= 1) f = 1;
    for (int i = 2; i <= n / 2; i++) {
        if (n % i == 0) { f = 1; break; }
    }
    if (f == 0) printf("%d is prime.\n", n);
    else printf("%d is not prime.\n", n);
}

void factorial(int n) {
    unsigned long long f = 1;
    if (n < 0) printf("Error.\n");
    else {
        for (int i = 1; i <= n; i++) f *= i;
        printf("Factorial: %llu\n", f);
    }
}

void prime_factors(int n) {
    printf("Prime factors: ");
    while (n % 2 == 0) { printf("2 "); n /= 2; }
    for (int i = 3; i <= sqrt(n); i += 2) {
        while (n % i == 0) { printf("%d ", i); n /= i; }
    }
    if (n > 2) printf("%d", n);
    printf("\n");
}

int main() {
    int n, choice;
    printf(" 1:Sqrt\n 2:Square\n 3:Cube\n 4:Prime\n 5:Fact\n 6:P-Factors\n");
    printf("Enter number: ");
    scanf("%d", &n);
    printf("Enter choice: ");
    scanf("%d", &choice);


    switch (choice) {
        case 1: printf("Sqrt: %.2f\n", sqrt(n)); break;
        case 2: printf("Square: %d\n", n * n); break;
        case 3: printf("Cube: %d\n", n * n * n); break;
        case 4: check_prime(n); break;
        case 5: factorial(n); break;
        case 6: prime_factors(n); break;
        default: printf("Invalid choice.\n");
    }
    return 0;
}
```

```
Overwriting a11.c
```

```
!gcc a11.c -o calculator -lm
!./calculator
```

```
 1:Sqrt
 2:Square
 3:Cube
 4:Prime
 5:Fact
```

```
 6:P-Factors
Enter number: 5
Enter choice: 5
Factorial: 120
```

Assignment 12

```
%%writefile a12.c

#include <stdio.h>

int find_scd(int a, int b) {
    int min = (a < b) ? a : b;
    for (int i = 2; i <= min; i++) {
        if (a % i == 0 && b % i == 0) return i;
    }
    return -1;
}

int find_gcd(int a, int b) {
    while (b != 0) {
        int temp = b;
        b = a % b;
        a = temp;
    }
    return a;
}

int main() {
    int n1, n2;
    printf("Enter two numbers: ");
    scanf("%d %d", &n1, &n2);

    int scd = find_scd(n1, n2);
    int gcd = find_gcd(n1, n2);

    if (scd != -1) printf("Smallest Common Divisor: %d\n", scd);
    else printf("No common divisor other than 1.\n");

    printf("Greatest Common Divisor: %d\n", gcd);

    return 0;
}
```

```
Overwriting a12.c
```

```
%%shell
gcc a12.c
./a.out
```

```
Enter two numbers: 75 125
Smallest Common Divisor: 5
Greatest Common Divisor: 25
```

Assignment 13

```
%%writefile a13.c

#include <stdio.h>

int main() {
    int n, rem;

    printf("Enter number: ");
    if (scanf("%d", &n) != 1) return 1;

    printf("Digits in reverse order: ");

    if (n == 0) printf("0");
    if (n < 0) n = -n;

    while (n > 0) {
        rem = n % 10;
        printf("%d ", rem);
        n = n / 10;
    }

    printf("\n");
```

```
    return 0;
}
```

```
Overwriting a13.c
```

```
%%shell
gcc a13.c
./a.out
```

```
Enter number: 13
Digits in reverse order: 3 1
```

Assignment 14

```
%%writefile a14.c

#include <stdio.h>

int main() {
    long long n;
    int dec = 0, base = 1, rem;

    printf("Enter a binary number: ");
    if (scanf("%lld", &n) != 1) return 1;

    long long temp = n;
    while (n > 0) {
        rem = n % 10;
        dec = dec + (rem * base);
        n = n / 10;
        base = base * 2;
    }

    printf("Decimal equivalent = %d\n", dec);

    return 0;
}
```

```
Writing a14.c
```

```
%%shell
gcc a14.c
./a.out
```

```
Enter a binary number: 1011
Decimal equivalent = 11
```

Assignment 15

```
%%writefile a15.c

#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main() {
    int n, i;

    printf("Enter number of random numbers needed: ");
    if (scanf("%d", &n) != 1) return 1;

    srand(time(NULL));

    printf("Pseudo Random Numbers:\n");
    for (i = 0; i < n; i++) {
        printf("%d\n", rand());
    }

    return 0;
}
```

```
Writing a15.c
```

```
%%shell
gcc a15.c
./a.out
```

```
Enter how many random numbers you want: 4
Pseudo Random Numbers:
262843796
892206681
1748483243
502150741
```

Assignment 16

```
%%writefile a16.c

#include <stdio.h>

int main() {
    int n, a[50], even[50], odd[50];
    int e_count = 0, o_count = 0;

    printf("Enter number of elements: ");
    if (scanf("%d", &n) != 1) return 1;

    printf("Enter integers: ");
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    for (int i = 0; i < n; i++) {
        if (a[i] % 2 == 0) {
            even[e_count++] = a[i];
        } else {
            odd[o_count++] = a[i];
        }
    }

    printf("\nEven numbers are: ");
    for (int i = 0; i < e_count; i++) {
        printf("%d ", even[i]);
    }

    printf("\nOdd numbers are: ");
    for (int i = 0; i < o_count; i++) {
        printf("%d ", odd[i]);
    }
    printf("\n");

    return 0;
}
```

```
Writing a16.c
```

```
%%shell
gcc a16.c
./a.out
```

```
Enter number of elements: 6
Enter integers: 3 45 6 2 4 77

Even numbers are: 6 2 4
Odd numbers are: 3 45 77
```

Assignment 17

```
%%writefile a17.c

#include <stdio.h>

int main() {
    int n, i;
    float x, sum, term;

    printf("Enter value of x (in radians): ");
    if (scanf("%f", &x) != 1) return 1;
    printf("Enter number of terms: ");
    scanf("%d", &n);

    sum = x;
    term = x;

    for (i = 1; i < n; i++) {
        term = -term * x * x / ((2 * i) * (2 * i + 1));
```

```
        sum += term;
    }

    printf("Sum of sine series = %f\n", sum);

    return 0;
}
```

```
Overwriting a17.c
```

```
%%shell
gcc a17.c
./a.out
```

```
Enter value of x (in radians): 1
Enter number of terms: 5
Sum of sine series = 0.841471
```

Assignment 18

```
%%writefile a18.c

#include <stdio.h>

int get_len(char s[]) {
    int i = 0;
    while (s[i] != '\0') i++;
    return i;
}

void get_rev(char s[], char r[]) {
    int len = get_len(s);
    for (int i = 0; i < len; i++) r[i] = s[len - 1 - i];
    r[len] = '\0';
}

int is_equal(char s1[], char s2[]) {
    int i = 0;
    while (s1[i] == s2[i]) {
        if (s1[i] == '\0') return 1;
        i++;
    }
    return 0;
}

int has_sub(char s[], char sub[]) {
    int i, j;
    for (i = 0; s[i] != '\0'; i++) {
        for (j = 0; sub[j] != '\0'; j++) {
            if (s[i + j] != sub[j]) break;
        }
        if (sub[j] == '\0') return 1;
    }
    return 0;
}

int main() {
    char s1[100], s2[100], sub[100], rev[100];

    printf("Enter main string: ");
    scanf("%s", s1);

    printf("Enter string for equality check: ");
    scanf("%s", s2);

    printf("Enter string to search (substring): ");
    scanf("%s", sub);

    printf("\n--- Results ---\n");
    printf("Length of main string: %d\n", get_len(s1));

    get_rev(s1, rev);
    printf("Reversed: %s\n", rev);

    if (is_equal(s1, rev)) printf("Status: Palindrome\n");
    else printf("Status: Not a Palindrome\n");

    if (is_equal(s1, s2)) printf("Equality: Match found\n");
    else printf("Equality: No match\n");
```

```
    if (has_sub(s1, sub)) printf("Substring: '%s' found in '%s'\n", sub, s1);
    else printf("Substring: Not found\n");

    return 0;
}
```

```
Overwriting a18.c
```

```
%%shell
gcc a18.c
./a.out
```

```
Enter main string: thisisthepartwhereisay
Enter string for equality check: abcdefghijklmnop
Enter string to search (substring): where

--- Results ---
Length of main string: 22
Reversed: yasierehwtrapehtsisiht
Status: Not a Palindrome
Equality: No match
Substring: 'where' found in 'thisisthepartwhereisay'
```

Assignment 19

```
%%writefile a19.c

#include <stdio.h>
#include <string.h>

struct EMPLOYEE {
    char name[50];
    char desig[50];
    char gender[10];
    char doj[20];
    float salary;
};

int main() {
    struct EMPLOYEE e[50];
    int n, male = 0, female = 0;

    printf("Enter number of employees: ");
    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        printf("\nDetails for Employee %d:\n", i + 1);
        printf("Name: "); scanf("%s", e[i].name);
        printf("Designation: "); scanf("%s", e[i].desig);
        printf("Gender (Male/Female): "); scanf("%s", e[i].gender);
        printf("Date of Joining (DD-MM-YYYY): "); scanf("%s", e[i].doj);
        printf("Salary: "); scanf("%f", &e[i].salary);

        if (strcmp(e[i].gender, "Male") == 0) male++;
        else if (strcmp(e[i].gender, "Female") == 0) female++;
    }

    printf("\n--- Organization Summary ---\n");
    printf("Total employees: %d\n", n);
    printf("Male: %d, Female: %d\n", male, female);

    printf("\nSalary > 10000: ");
    for (int i = 0; i < n; i++) {
        if (e[i].salary > 10000) printf("%s ", e[i].name);
    }

    printf("\nAsst Managers: ");
    for (int i = 0; i < n; i++) {
        if (strcmp(e[i].desig, "AsstManager") == 0) printf("%s ", e[i].name);
    }
    printf("\n");

    return 0;
}
```

```
Overwriting a19.c
```

```
%%shell
gcc a19.c
./a.out
```

```
Enter number of employees: 3

Details for Employee 1:
Name: abc
Designation: AsstManager
Gender (Male/Female): Female
Date of Joining (DD-MM-YYYY): 12-04-2001
Salary: 60000

Details for Employee 2:
Name: def
Designation: GenManager
Gender (Male/Female): Male
Date of Joining (DD-MM-YYYY): 31-07-2020
Salary: 17000

Details for Employee 3:
Name: ghi
Designation: Intern
Gender (Male/Female): Male
Date of Joining (DD-MM-YYYY): 23-02-2025
Salary: 10000

--- Organization Summary ---
Total employees: 3
Male: 2, Female: 1

Salary > 10000: abc def
Asst Managers: abc
```

Assignment 20

```
%%writefile a20.c

#include <stdio.h>

void swap_value(int a, int b) {
    int temp = a;
    a = b;
    b = temp;
}

void swap_reference(int *x, int *y) {
    int temp = *x;
    *x = *y;
    *y = temp;
}

int main() {
    int n1, n2;

    printf("Enter two numbers: ");
    scanf("%d %d", &n1, &n2);

    printf("\n--- Call by Value ---\n");
    swap_value(n1, n2);
    printf("Swapped without pointers: n1 = %d, n2 = %d\n", n1, n2);

    printf("\n--- Call by Reference ---\n");
    swap_reference(&n1, &n2);
    printf("Swapped with pointers: n1 = %d, n2 = %d\n", n1, n2);

    return 0;
}
```

```
Overwriting a20.c
```

```
%%shell
gcc a20.c
./a.out
```

```
Enter two numbers: 23 45

--- Call by Value ---
Swapped without pointers: n1 = 23, n2 = 45

--- Call by Reference ---
Swapped with pointers: n1 = 45, n2 = 23
```

Assignment 21

```
%%writefile a21.c

#include <stdio.h>
#include <stdlib.h>

int main() {
    FILE *fs, *fd;
    char ch;
    char s_name[20], d_name[20];

    printf("Enter source file name: ");
    scanf("%s", s_name);

    fs = fopen(s_name, "r");
    if (fs == NULL) {
        printf("Error: Cannot open source file.\n");
        return 1;
    }

    printf("Enter destination file name: ");
    scanf("%s", d_name);

    fd = fopen(d_name, "w");
    if (fd == NULL) {
        printf("Error: Cannot open destination file.\n");
        fclose(fs);
        return 1;
    }

    while ((ch = fgetc(fs)) != EOF) {
        fputc(ch, fd);
    }

    printf("File copied successfully.\n");

    fclose(fs);
    fclose(fd);

    return 0;
}
```

```
Overwriting a21.c
```

```
%%writefile source.txt
Hello
Welcome to C Programming
```

```
Overwriting source.txt
```

```
%%writefile dest.txt
```

```
Writing dest.txt
```

```
%%shell
gcc a21.c
./a.out
```

```
Enter source file name: source.txt
Enter destination file name: dest.txt
File copied successfully.
```

```
!cat dest.txt
```

```
Hello
Welcome to C Programming
```

Assignment 22

```
%%writefile a22.c

#include <stdio.h>
#include <string.h>

int main() {
    char s1[100], s2[100], temp[100];
    int choice;

    printf("Enter first string: ");
    scanf("%s", s1);
```

```
    printf("Enter second string: ");
    scanf("%s", s2);

    printf("\n--- MENU ---\n");
    printf("1. Length\n2. Copy\n3. Concatenation\n4. Comparison\n5. Reverse\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            printf("Length of S1: %ld, S2: %ld\n", strlen(s1), strlen(s2));
            break;
        case 2:
            strcpy(temp, s1);
            printf("S1 copied to temp. Temp = %s\n", temp);
            break;
        case 3:
            strcat(s1, s2);
            printf("Concatenated string: %s\n", s1);
            break;
        case 4:
            if (strcmp(s1, s2) == 0) printf("Strings are equal.\n");
            else printf("Strings are not equal.\n");
            break;
        case 5:
            printf("Reversed S1: ");
            for (int i = strlen(s1) - 1; i >= 0; i--) printf("%c", s1[i]);
            printf("\n");
            break;
        default:
            printf("Invalid choice.\n");
    }

    return 0;
}
```

```
Overwriting a22.c
```

```
%%shell
gcc a22.c
./a.out
```

```
Enter first string: abcdefghijkl
Enter second string: mnopqrstuvwxyz

--- MENU ---
1. Length
2. Copy
3. Concatenation
4. Comparison
5. Reverse
Enter choice: 3
Concatenated string: abcdefghijklmnopqrstuvwxyz
```

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